

AB \ CD	00	01	11	10
00			0	
01	0		0	0
11	0	0	0	0
10			0	

$$F = CD + AB + BD'$$

$$(F')' = F = (CD + AB + BD')$$

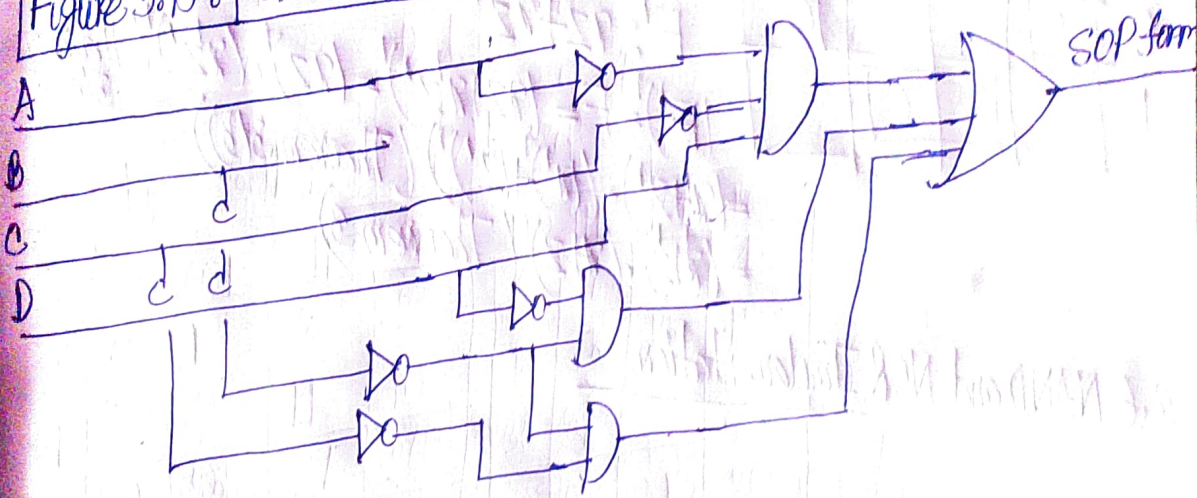
$$= (CD)' \cdot (AB)' \cdot (BD')' \quad \text{(Extended De Morgan's Theorem)}$$

$$= (C'D') \cdot (A'B') \cdot (B+D) \quad \text{(Theorem 5(b))}$$

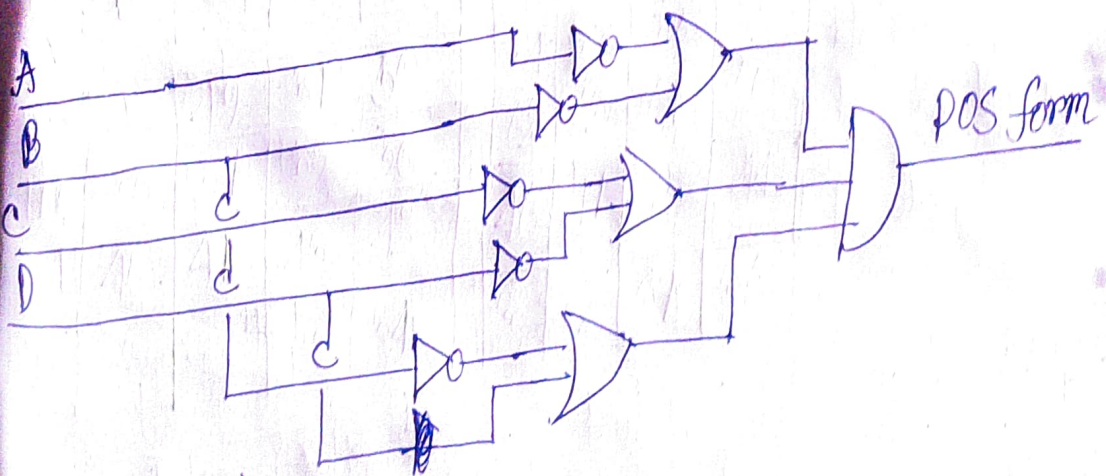
$$= (C'D') \cdot (A'B') \cdot (B+D) \quad \text{(Theorem 3)}$$

(Product of Sum Form)

Figure 3.15: Gate implementation of the function of Example 3-8



(a)



(b)